FormuLab User Guide

Local-first formulation research workbench

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FormuLab — literature-driven chemical formulation discovery, laboratory and stability tracking, regulatory dossiers, and cost optimization, for Kenya/EAC R&D teams. Local-first: your projects live on your own machine, never on a server you don't control.

This is the walkthrough for the full R&D workflow: generate a starting idea, build a formula, save versions, test it in the lab and on the shelf, clear it for the markets you sell into, price it against real materials, and compare. Each section links to the document that covers the topic in depth; this page is the map, not a replacement for those.

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0. Navigating FormuLab

[Screenshot not yet captured: Home]

The sidebar's Workspaces section is the primary navigation: Home (a real dashboard — recent projects, activity, open lab work, upcoming stability samples, pending approvals), Projects (every formulation project), Formulation (the grid, versions, cost, compatibility, safety, packaging — sections 2–13 below), Laboratory (trials, test definitions, corrective actions — sections 16–19), Stability (section 18), Optimization (section 14), Regulatory (section 20), Approval (sections 5, 12), Reports (a navigation shell over each module's existing export), Data Exchange (section 23 — bulk CSV/Excel import/export across every module), and Administration (materials/ suppliers/packaging/factory profiles, plus links to regulatory rules, approval policies, the Data Exchange Center and app settings).

Opening a project from Projects or Home carries it with you as a ?project= link — Laboratory, Stability, Optimization, Regulatory and Approval all open already scoped to that project; a blocker inside Approval that names a specific module opens that module directly, on the right formula version. See WORKSPACES.md and NAVIGATION_AND_CONTEXT.md for the full model. This replaced a single crowded Formula Builder page that carried every one of these as a horizontal tab; nothing described in this guide was removed or rewritten — every section below still describes real, working functionality, just reached from its own workspace rather than a tab.

0a. Getting help in the app

Every workspace above has a small Help button (bottom-right corner) that opens a side panel explaining the current page — summary, purpose, prerequisites, a quick-start list, its sections, warnings, role notes, and known limitations, with clickable related topics and glossary terms that switch the panel in place (never leaving your page or losing your work). A page reached through a nested URL (e.g. a Settings section, or a live session) shows its parent page's help when it has no dedicated topic of its own.

[Screenshot not yet captured: The page Help panel]

Press Ctrl/Cmd+/ (or run Search help from the command palette, Ctrl/Cmd+K) to open the global Help Center — a full-text search across every topic's title, summary, keywords, and quick-start text, plus every glossary term, regardless of which page you're on. Selecting a result opens the same Help panel; searching help never navigates you away from what you're doing. A handful of pages (an old redirect, the retained legacy Formula Builder, the 404 page) intentionally have no help topic — the Help button simply doesn't appear there.

[Screenshot not yet captured: The Help Center]

Laboratory's help topic covers the Session 1A test-standards/methods feature — selecting a standard, primary vs. alternative methods, internal methods, and the official-standard copyright notice — reachable from Laboratory's own Help button or by searching "standard" or "method" in the Help Center.

A small i icon next to some fields (Formulation's total/q.s. badge and cost snapshot, Laboratory's method status badges, Design of Experiments' factors/responses, Stability's conditions/time points) opens a short explanation on hover or keyboard focus, with a "Learn more" link into that page's own Help panel where relevant.

!An InfoTooltip open

When a button is greyed out (Laboratory's Make primary, Approval's Approve, a Dossier's Export PDF/DOCX, Data Exchange's Commit import, Formulation's Save cost snapshot), the reason is written directly beneath it — what's blocking it, which role can act if it's a permission issue, and whether you can resolve it yourself. A disabled button never reacts to a click; the explanation is not a hint to try anyway.

[Screenshot not yet captured: A disabled-action explanation]

Three short guided tours — Formulation (the session composer where a formulation is generated), Design of Experiments, and Dossiers — walk through each workspace's real controls with a spotlight and a small step card. Every tour is optional and skippable at any point, supports Next/Back/Skip/Finish, closes on Escape, and returns keyboard focus to wherever you were when you started it. Start one from that page's Help panel ("Start tour"), or from the one-time onboarding prompt shown the first time you open the app (dismiss it with the x or "Maybe later" — it never reappears once dismissed or once you've picked a tour). A tour never edits any project data; it only highlights.

[Screenshot not yet captured: The first-use onboarding prompt]

[Screenshot not yet captured: A guided tour step]

This guide itself is one Markdown source, rendered three ways: in-app (Help Center -> Open the full guide, or the command palette), as a PDF (docs/generated/FormuLab-User-Guide.pdf), and as a DOCX (docs/generated/FormuLab-User-Guide.docx) — see §29. English-only for v1, the same locale policy every screenshot in this guide follows.

0b. Generate a starting formulation (the composer)

[Screenshot not yet captured: Formulation generation]

Sessions (the sidebar's pinned bottom section, or Ctrl/Cmd+B's default landing page) opens the composer at /live — FormuLab's front door for a brand-new idea, distinct from the Formulation project workspace described from §1 onward.

- Describe the target product in plain language (e.g. "a sulfate-free gentle shampoo"), optionally narrow the category and market, then click Generate. This runs the real discovery pipeline — retrieve open-access literature, extract cited compositions, synthesize a brief, optimize, and report — never a single unchecked model guess. A target FormuLab's safety classifier flags as hazardous/regulated/medical asks for a named human's acknowledgement before it proceeds; a prohibited target is refused outright. See §11.

• The result streams back as one or more numbered candidates (V1, V2, V3, …), each a citable, printable formulation card.

!Candidate versions

• Card shows the generated write-up and a cost estimate; Edit formula opens that exact candidate in the same grid §2 uses, so you can adjust it immediately — each candidate keeps its own independent, unsaved edits; editing V2 never touches V1 or V3.

!Edit formula, function totals and warnings

• The same function totals and validation findings from §2 apply here — a badge like "anionic surfactant: 18%" is always the real formula percentage, never zeroed just because a material's active-matter content is unrecorded; an incomplete active-matter figure is reported as its own honest (active data incomplete) note, not folded into the percentage.

Note: A generated candidate is not automatically a project. The composer and the Formulation project workspace do not currently bridge automatically — there is no "save this candidate as a project version" button. If a candidate looks promising, start a real project (§1) and rebuild it there, using the candidate as your reference; only a project version gets save-history, approval, laboratory/stability/regulatory tracking, and dossiers.

Every past run appears in the sidebar's Sessions list (newest first, up to 8 shown, View all for the rest) — click one to reopen it read-only (no new model call); delete one with the trash icon next to it. A session never appears there unless it actually produced at least one card — a refused or failed run is never listed as if it succeeded.

1. Create a formula project

[Screenshot not yet captured: Projects]

Projects -> New project in the sidebar (or Home -> Recent projects once you have one).

• Pick a product family from the Kenya catalog (55 families across 17 domains). This determines which packaging SKUs are offered and which structural template pre-populates the grid.
• Pick one or more packaging SKUs the formula is meant to fill.
• Enter project name, optional project code (generated if left blank), product brief, target market (Kenya or EAC), target claims, and target batch size.
• Save. The project persists to data/formulations/<id>/ — closing the app and reopening it later reopens the same project, not a blank grid.

See FORMULA_BUILDER.md for the full field list and the templates for all 35 distinct product types.

2. Build the formula

The grid is the daily surface — not the chat thread. Add lines, set phase, material, percentage and active-matter %; drag to reorder; move a line between phases by editing its phase cell.

• Water q.s.: mark one line q.s. and it fills automatically to 100% of whatever the other lines leave, and never goes negative. Convert it to a fixed percentage, or a fixed water line back to q.s., at any time.
• Validation runs continuously: formula total vs. 100%, technical maxima, missing material references, invalid batch size, and more, at info / warning / error / blocking severity. Blocking findings are shown at the line and in a formula-level summary.
• Functional-group totals (total anionic surfactant %, total preservative %, ...) report incomplete rather than silently treating a missing active-matter figure as zero.
• Undo/redo, autosave (writes the working draft only), and a visible unsaved-changes indicator are all live controls, not decoration.

Full control reference, keyboard shortcuts, and paste-from-Excel behaviour: FORMULA_BUILDER.md.

Precision rules for every number shown: PRECISION_POLICY.md.

3. Save a version

Save version promotes the current working draft to an immutable, numbered version and requires a change reason. A saved version is never edited in place — editing it again means editing a new draft derived from it, which saves as a new version with the old one recorded as its parent.

Every version freezes a snapshot of totals, validation results, and intent (market, claims, SKUs) as they stood at save time, so a later edit to the project brief cannot rewrite what a version says it was.

Full model: FORMULA_VERSIONING.md.

4. Compare versions

Open the Compare view and pick two versions. It reports, per line: added, removed, changed or unchanged, with field-level detail (percentage, batch quantity, supplier, price, currency, function, phase, INCI, evidence origin), plus totals, active matter, and status changes — as a diff, not prose. Any inferred performance impact is explicitly labelled estimated / requires laboratory confirmation; nothing here claims a measured result.

5. Approval

Open the Approval workspace (/approval) for the selected project. No automated actor — agent, system process, or import — can set pilot_approved or production_approved, regardless of what a model concluded or a spreadsheet claimed. Approval is a named human action with a signed record and an audit entry. Clones and restores of an approved version always start unapproved. Details and the bypass tests: FORMULA_VERSIONING.md.

6. Raw materials and suppliers

Administration -> Materials, suppliers, packaging & factory profiles in the sidebar. Create a material with whatever is known today — internal code and function are enough to start; every other field is explicitly missing, unknown, not_applicable or not_verified rather than blank-meaning-zero. Attach one or more suppliers to a material (one trade name is not assumed to map to one supplier).

Search and filter by function, ionic character, supplier, country, stock status, and data-completeness flags (has SDS, has price, has density, ...).

Full field list and data-state model: RAW_MATERIALS.md.

7. Prices, landed cost, and inventory

Prices are append-only history, not a single current field — a price change today never rewrites what a formula cost in March. Each price record can carry freight, insurance, duty, tax and other landed-cost components; the engine supports per-kg, per-shipment, percentage-of-goods and fixed-amount allocation.

Inventory records (lot, quantity, reserved/available, expiry, COA and quarantine status) support stock-awareness in costing without pretending to be a full ERP module.

Import materials, suppliers, material-supplier links, prices, inventory, packaging components, packaging BOMs and factory cost profiles from CSV or Excel (.xlsx): preview rows, see row-level errors and warnings separately, and choose whether to commit a partial import. Import is idempotent on the stable internal code, so re-importing the same file does not create duplicates. Spreadsheet formula injection is stripped on import and on export; macro-bearing or otherwise unsupported binary content is rejected rather than executed. Downloadable .xlsx templates exist for every importable collection. See IMPORT_EXPORT.md.

Materials also has editor screens for supplier detail (contact, Incoterm, payment terms, lead time, MOQ notes, approved-supplier and quality status, linked materials and price history), packaging components and packaging BOMs (component list, add/remove/reorder, quantity per SKU, waste factor, carton and shrink-wrap allocation, total packaging cost), and factory cost profiles (create, edit, clone, activate/deactivate, effective date, utility and labour rates, QC allocation, waste rate, overhead basis). Demonstration figures on a factory profile stay marked example_only / not_verified until someone replaces them with real factory data.

8. Cost a formula

Open the Cost tab in the Formulation workspace. It calculates, as distinct layers — never merged into one number — raw-material cost, landed cost, packaging cost, labour, utilities, QC, waste and factory overhead, then rolls up to cost per kg, per litre (when density is known), per batch and per packaging SKU (per sachet, bottle, drum, ...).

Currency conversion uses exchange rates you enter and date — FormuLab never calls an external rate API. Factory cost profiles (electricity, water, labour rate, overhead basis) are editable per factory and marked verified / not_verified / example_only so a demonstration number can never be mistaken for real production data.

Full model: COST_ENGINE.md.

9. Save a cost snapshot

Save cost snapshot freezes the exact price records, exchange rates, packaging costs and factory profile used, dated. Updating a material's current price afterwards does not change that snapshot — a new snapshot must be created explicitly to see the effect of the new price.

Comparing two versions' cost snapshots attributes the difference to formula change, price change, exchange-rate change, packaging change or factory-cost change, and reports missing-data impact as its own category rather than folding it into one of the others.

10. Compatibility

Open the Compatibility tab in the Formulation workspace. A deterministic rule engine — not the LLM — checks the formula's materials, functions, ionic character, concentrations, target pH, process temperature, addition order and packaging against a hand-maintained, versioned rule set (anionic/cationic, QAC/anionic, chlorhexidine, hypochlorite interactions, oxidizer/reducer, carbomer, packaging incompatibilities, and more). Findings are info / warning / error / blocking; missing data produces an explicit "unknown — data missing" finding, never a false pass. Findings are snapshotted onto a saved version and re-run whenever the draft changes. Manage rules lets you create, edit, deprecate and import/export rules as JSON or Excel. Full model: COMPATIBILITY_ENGINE.md.

11. Safety

Open the Safety tab in the Formulation workspace. The product is classified deterministically (ordinary consumer / industrial / hazardous lawful / regulated disinfectant / medical / restricted / prohibited / human-review-required), and a versioned rule set checks for known hazard interactions, corrosivity, pH extremes, sensitizer and acute-toxicity thresholds, and more. A blocking finding cannot be dismissed — resolving one requires a named reviewer, a reason, a date and an audit record; the LLM can never resolve or approve one on its own. The same deterministic classification also gates AI formulation requests before literature discovery runs: a prohibited target is refused outright, a hazardous/regulated/medical one requires a named human's acknowledgement before generation proceeds. Full model: SAFETY_ENGINE.md.

12. Approval readiness

A formula cannot reach pilot_approved or production_approved while any blocking validation error, blocking compatibility finding, blocking safety finding, or unresolved mandatory human review is open — the UI names every blocker. See APPROVAL_READINESS.md.

13. Lifecycle, variants and exports

Saved versions are immutable but not static: retire or reject a version with a reason, reopen a rejected one into a new draft, and create a named variant from any saved version to branch exploration without disturbing the parent line. None of this — retire, reopen, clone, or variant creation — ever inherits or grants production approval.

Export a selected version as a JSON formulation package, CSV or Excel formula sheet, cost-snapshot JSON, packaging-BOM JSON, or draft ERP BOM/ recipe CSV. Every export carries formula ID, version ID, version label, schema version, export timestamp, approval status, cost-snapshot ID, and target family/SKUs; a non-approved formula is stamped R&D DRAFT — NOT PRODUCTION APPROVED on every export format.

14. Advanced optimization

Open the Optimization workspace (/optimization) for the selected project — distinct from the standalone what-if calculator linked from it, which is not project-bound. Unlike the plain cost-minimizing Formulation Optimizer, this is a real constraint-solving workspace: pick candidate materials, add functional-group constraints (e.g. "at least 15% anionic surfactant") — soft or hard, with a penalty weight and allowed deviation for a soft one — property targets (calculated for real where the platform honestly can: active matter, total solids, several named actives; a laboratory_required property is never given a fabricated value), an optional cost ceiling, and one or more objectives (cost, supply risk, carbon score, stock utilization, evidence confidence, and graded compatibility/ safety risk — weighted together, or lexicographic priority). Every candidate pair is automatically checked against the real Compatibility and Safety engines before the solve, so the optimizer can never select a combination those engines flag as blocking. Run — a solve that had to relax a soft constraint reports feasible_with_penalties, distinct from a clean optimal, with each soft constraint's requested target, achieved value and deviation shown. An infeasible run explains why in plain language with suggested next steps, not just "infeasible". Apply to draft never overwrites the saved version it started from — it creates a new working draft, and the run is remembered so approval readiness can later verify it was genuinely usable. Full model: ADVANCED_OPTIMIZER.md.

Scenarios — the same tab's Scenarios section lets you name and save a "what if" (its full candidate/constraint/objective selection and a frozen price/inventory snapshot), reload it later, clone it, rename it, retire it, or restore a retired one as a new scenario. Load one of the 31 seeded Kenya product-family structural profiles (apply only what's missing, merge it in, or replace your current selection — the last asks for confirmation first) as a starting point; every profile is explicitly not_verified and needs chemist review, never an approved recipe. Select two or more runs (from one scenario's history or several scenarios) and Compare to see cost, risk, soft violations, stock utilization and solve time side by side, with the lowest/highest of each highlighted — never a single fabricated "best overall" score. Full model: OPTIMIZATION_SCENARIOS.md.

15. Material substitution

Click the replace-material icon on any formula line to open scored, ranked candidates for that material — scored on real data (function match, active-matter equivalence, ionic character, pH/HLB similarity, regulatory status, available stock, landed cost, and a live compatibility/safety re-check), never by name similarity. A candidate that would introduce a blocking finding is still shown, sorted last, so you can see why it ranked where it did. Applying a candidate uses the active-equivalent percentage (10% of a 70%-active material needs 20% of a 35%-active replacement to contribute the same active matter) and, like an optimizer result, only ever creates a new working draft.

System substitution — check additional formula lines in the same dialog to replace several lines with a whole new material system at once (a surfactant blend, a preservation system, a thickener + neutralizer, ...). Pick which functions to preserve, generate candidate systems (by real function coverage, never name similarity, within configurable candidate limits), then Evaluate through optimizer to route every proposal through the actual Advanced Optimizer and score the results. A rejected combination shows why; an infeasible one shows its structured cause. Applying a system replaces every selected line with the new materials in a new working draft. Full model: MATERIAL_SUBSTITUTION.md, SYSTEM_SUBSTITUTION.md.

16. Laboratory trials

Open the Laboratory workspace (/laboratory)'s Trials section. Create a trial from the current working draft (or a saved version, if one is selected) — it freezes its own formula snapshot, so later formula edits never change what the trial recorded. Move it through its lifecycle (planned -> materials prepared -> in progress -> awaiting results -> completed/failed -> archived) with the status buttons; an invalid move is refused, and only a human can mark a trial completed.

- Material weighing: enter each material's actual weight; the computed deviation and batch-level variance appear immediately. A material with no actual weight yet shows "not entered," never a zero.
- Process execution: add process steps, record actual temperature/pH/ duration against the planned range.
- Observations & deviations: log an observation, file a deviation (minor/major/critical), resolve it or accept it with a written justification, and open a corrective action directly from an unresolved one. A critical open deviation blocks the trial from being marked complete.
- Test results: enter replicate values for any active numeric test definition; mean, standard deviation and pass/fail compute automatically from the test's own logic. Click View history on any result to open the dedicated result history browser — every revision, retest, override and attachment replacement, plus a side-by-side comparison of any two revisions. Click Test applicability to see which test definitions are included/excluded for this trial's product family and packaging, and exactly why.

Select two or more trials (checkbox in the list) and Compare selected to see material-usage, deviation and pass/fail counts side by side — a deterministic comparison, never an inferred "why."

Export (per trial): JSON package, batch sheet (CSV), weighing sheet (CSV), process sheet (Excel), test-results report (Excel), corrective- actions report (CSV), and a draft ERP lab-result CSV — every export watermarked R&D DRAFT — NOT PRODUCTION APPROVED unless the source formula is genuinely production_approved. Full model: LABORATORY_TRIALS.md, TRIAL_EXECUTION.md, TRIAL_COMPARISON.md.

17. Test definitions

Open the Laboratory workspace's Test Definitions section (also reachable from Administration, since this catalog is global rather than per-project) to manage the reusable test-definition catalog shared by trials and stability studies — 27 structural templates ship seeded (pH, viscosity, density, foam, microbiology, preservative challenge, and more), all explicitly not_verified until a chemist attaches their own method reference and marks one verified. Edit result type, unit, min/max, pass/fail rule, critical flag and active status inline. Full model: TEST_DEFINITIONS.md.

17a. Test standards and methods

Each test definition can now carry its own configurable standard(s) and method, independent of every other test — changing pH's standard never touches viscosity's. Open a definition's Method button (next to it in Test Definitions) to manage this.

[Screenshot not yet captured: Laboratory test method drawer]

- Selecting a method: a test may have one primary method and any number of alternative methods, each linking a LaboratoryStandard (code, title, issuing organization, edition/revision, status) to procedural detail (equipment, reagents, sample prep, instrument settings, steps, calculations, acceptance criteria, safety, waste

disposal, and more — see the Method drawer's own sections).
• Primary vs. alternative: only chemist/quality/administrator roles may promote an alternative to primary or create a new method; any role may view. Promoting a method demotes the previous primary to alternative — a test is never left with two primaries.
• Internal methods: an authorized user may create a FormuLab-internal method (no external standard body) directly from the drawer; it is marked internal and behaves like any other standard for assignment purposes.
• Status and revision: a standard is draft, active, superseded, or internal. Selecting a superseded standard as primary requires explicitly checking an acknowledgement box first — it is never silently selectable.
• Detailed instructions: the Method drawer's sections (overview, scope, equipment, reagents, sample prep, instrument setup, conditioning, procedure, calculations, acceptance criteria, result interpretation, troubleshooting, repeat-test conditions, safety, waste disposal, related tests, alternative standards, revision/source) render whatever has actually been entered; an unfilled section says so honestly rather than inventing content.
• Historical snapshots: recording a test result can capture an immutable copy of the method actually used (standard code/edition/ revision, method id/version, instrument settings, unit, acceptance criteria). A later edit to the standard or method never changes an already-recorded result's snapshot.
• Legacy references: a test definition's older free-text methodReference field (e.g. "ISO 4316") is never auto-converted into a structured standard — it stays visible as an unresolved legacy reference until a chemist creates the real structured standard/method themselves.
• Copyright notice: FormuLab never reproduces or invents the full text of a copyrighted standard (ISO/ASTM/EN/DIN/AOAC/USP/EP/BS/...). The drawer always shows a notice that its summaries and internal procedures do not replace the official licensed standard — consult the current official publication before regulated testing.

Full model: packages/shared/src/schemas/laboratoryStandards.ts, packages/shared/src/engine/laboratoryStandards.ts.

18. Stability studies

Open the Stability workspace (/stability). Create a study against the current working draft or a saved version (frozen formula + packaging snapshot), pick which seeded storage conditions, time points and test definitions apply — these are configurable starting examples, never a claim of what any regulator requires — and move it to active (this sets its start date). Click Test applicability before creating the study to see the same Included/Excluded explorer Trials uses, scoped to this study's product family, packaging, conditions and time points; selecting a test the explorer marks excluded prompts for a reviewer and reason, both recorded permanently in the study's requirement snapshot. Generate samples creates one pull-point sample per condition × time point × replicate, each with a deterministically computed due date.

Record a result for a due sample — click View history on it for the same result history browser Trials uses. An out-of-range numeric result automatically opens a stability failure — critical when the test is flagged critical. Trends shows one small chart per condition × test metric (change from initial, rate per day, min/ max/mean); a projection toward a limit only ever appears once enough real data exists, and is always labelled "experimental estimate — not validated — human review required," never a shelf-life claim. Resolve a failure, open a corrective action from it, or create a draft formula from that action — same as the Trials workspace.

Export: protocol (JSON), sample plan (CSV), time-point report (Excel), summary report (Excel), test-results report (Excel), corrective-actions report (CSV), and a draft ERP lab-result CSV. Full model: STABILITY_STUDIES.md, STABILITY_TRENDS.md.

19. Corrective actions

Open the Laboratory workspace's Corrective Actions section for the cross-cutting list of every action opened against a trial deviation or stability failure for this project (a trial/study's own workspace also shows its actions inline). Move one through start progress -> mark complete -> verify effective/ ineffective; effective/ineffective only exist after that verification step, never set directly. Create draft branches a new working draft from the action's source formula version — never inheriting approval, and never mutating the version it branched from. Export the whole list as CSV. Full model: CORRECTIVE_ACTIONS.md.

20. Regulatory (Kenya/EAC)

Open the Regulatory workspace (/regulatory). First pick a saved formula version — findings, confirmations, and reviews are all recorded against this exact version, never the working draft — then a jurisdiction (Kenya, Uganda, Tanzania, Rwanda, Burundi, South Sudan, or the EAC regional bloc), a packaging SKU (where relevant), and the reviewer role you're acting as. The classification card shows this project's deterministic regulatory category with its reasoning, never a model's guess. Click Evaluate to run every applicable rule (the jurisdiction's own plus any active EAC rule) against the current formula and see the resulting findings — compliant, non-compliant, missing data, or human review required, each with its reason and, where relevant, controls to confirm a requirement is satisfied or evidence has been provided. These confirmations are now persisted (not a session-local checkbox) — they survive a reload and can be revoked with a reason.

The Rules section lists every rule for the selected jurisdiction — 17 seed placeholders ship across all seven jurisdictions, every one explicitly not_verified pending a qualified regulatory reviewer's confirmation. Create, edit, activate/deactivate, deprecate, verify, reject a verification, or supersede a rule (edits, deprecations, and verification decisions require a reason, recorded in the rule's own revision history); import/export the rule set as JSON, CSV, or Excel — every import previews the parsed rows before you commit.

The Reviews section records a human's regulatory sign-off (reviewer name, outcome, notes) bound to the exact version/jurisdiction/SKU selected above, and lets you revoke a review with a reason. A separate "declare equivalence" control lets an authorized human explicitly permit reusing one version's review for another version, scoped to jurisdiction and packaging SKU — never assumed automatically.

Turning on any of the Approval workspace's regulatory-gate toggles (classification completed, no blocking finding, mandatory documents/ evidence/claims reviewed, human review completed) folds these facts into that formula's readiness check, the same way the cost-snapshot gate already works — and, when a policy is configured to require more than the primary market, across every required jurisdiction at once, not just the first one. Full model: REGULATORY_ENGINE.md, REGULATORY_CLASSIFICATION.md, REGULATORY_RULES.md, EAC_MARKET_PROFILES.md, REGULATORY_REVIEWS.md, REGULATORY_EVIDENCE_CONFIRMATIONS.md, REGULATORY_MULTI_MARKET_APPROVAL.md, REGULATORY_RULE_VERIFICATION.md.

21. Dossiers and Claims & Labels

Open the Dossiers workspace (/dossiers) to build a per-version, per- jurisdiction regulatory dossier: create one against a real saved formula version, then work through Requirements, the Evidence Library (with suggested evidence pulled from your own raw-material documents, lab trials, stability results and regulatory reviews), the live Evidence Matrix (filterable, exportable), Reviews and Submissions. Evidence can be replaced (never edited in place) with a full revision chain kept visible.

[Screenshot not yet captured: Dossier readiness and export]

Once enough evidence is accepted, the readiness summary shows exactly how many mandatory requirements are satisfied and what's still missing — never a single pass/fail flag standing in for the real detail. Export PDF / Export DOCX (Evidence Library section) render a real, watermarked document from the current evidence matrix — see §21a. Exporting never approves anything by itself: approval only happens in the Approval workspace (§5/§12).

21a. Dossier PDF/DOCX export

A Dossier's Export PDF/Export DOCX buttons (authorized regulatory/quality/administrator roles only — shown disabled with the exact reason for anyone else, never hidden) generate a real, deterministic document straight from pdf-lib/docx — the same two libraries this guide's own PDF/DOCX exports use (§0a) — never an HTML/DOM screenshot. Every export is persisted as a dated export-history record and an audit event, whether it succeeded or failed, and is watermarked R&D DRAFT — NOT PRODUCTION APPROVED unless the source formula version is genuinely production_approved.

Open the Claims & Labels workspace (/claims-labels) to manage product claims and product labels for the same formula version. A claim starts as a draft with an auto-classified category (30 categories, from performance to medical to eco); link it to evidence already in a dossier (never a duplicated file) and record a formal review once evidence is accepted — only that review, from an authorized regulatory/quality/administrator role, changes its true status. A label is one jurisdiction/language combination for one formula version — fill in its structured content sections (Front/Back/Side panel, Ingredients, Directions, Warnings, Claims, Manufacturer, Codes), upload artwork and get it approved, then run the Consistency check to catch a label pointing at the wrong formula version, an incomplete ingredient declaration, or a prohibited claim printed on the label. Both claims and labels support JSON/CSV/Excel import/export with a preview step; imported records always start as drafts, never auto-verified or auto-approved.

Turning on any of the Approval workspace's claims/label-gate toggles (all claims reviewed, no prohibited/unsupported claims, label review complete, artwork approved, formula-label consistency, all required languages reviewed) folds these facts into that formula's readiness check the same way the dossier gates already do. Full model: PRODUCT_CLAIMS.md, CLAIM_EVIDENCE.md, CLAIM_REVIEWS.md, PRODUCT_LABELS.md, LABEL_CONTENT.md, LABEL_ARTWORK.md, LABEL_REVIEWS.md, FORMULA_LABEL_CONSISTENCY.md, CLAIMS_LABEL_READINESS.md.

22. Design of Experiments

Open the Design of Experiments workspace (/doe) to plan and run a statistically valid formulation/process experiment. Create a study through the wizard: pick a real saved formula version as its baseline, define factors (formula materials, mixing speed/time, temperature, pH, or a custom process parameter — each with a low/center/high range or a categorical level list), add any hard/soft constraints, define responses and their objectives (maximize/minimize/target/within-range), pick a design type (full/fractional/two-level factorial, Plackett-Burman, central composite, Box-Behnken, Latin hypercube, mixture simplex-lattice, or a hand-built custom design), and generate — the app randomizes the run order and shows real diagnostics (duplicate/balance/orthogonality/condition- number checks) before you commit.

Work through the generated Runs: record each response's observed value per run (or mark it missing/invalid/excluded with a reason — never silently treated as zero), and generate or link a real Laboratory trial directly from a run. Once enough data is in, run a statistical analysis — the app fits a real OLS model (main effects, factorial with interactions, quadratic response-surface, or a mixture blending model) to what was actually recorded, and shows the coefficients, ANOVA, fit quality, and residual charts. It will suggest — never auto-exclude — a run that looks like an outlier; you decide. Generate candidates from a finished analysis to see a ranked, desirability-scored list of promising factor settings, then apply the best one to your current working draft (it never overwrites a saved version) and save it as a new version through the normal Formulation workflow when you're satisfied.

Full model: DESIGN_OF_EXPERIMENTS.md and its companion documents (DOE_STUDIES.md, DOE_FACTORS_AND_CONSTRAINTS.md, DOE_DESIGN_GENERATION.md, DOE_RESPONSES.md, DOE_RUNS_AND_LABORATORY.md, DOE_STATISTICAL_ANALYSIS.md, DOE_CANDIDATES.md, DOE_OPTIMIZATION_INTEGRATION.md).

23. Data Exchange Center

Open the Data Exchange Center (/data-exchange, sidebar between Reports and Administration) to bulk import or export structured data — materials, suppliers, prices, formulas, lab/stability results, regulatory rules, dossier evidence, claims, label content, DOE data, Reverse Formulation studies, and more — as CSV or real Excel files, across 41 templates (the exact, test-enforced count — see packages/shared/src/engine/dataExchangeRegistry.ts).

For any template's card in the Template Library: download a Blank file (header row only) to fill in from scratch, an Example file (a few synthetic, clearly-TEST--prefixed rows) to see the expected shape, or the current data already in that collection to review or hand off. Excel downloads include dropdown validation on every enum column and a Field Documentation sheet, so you don't need this guide open while filling one in.

To import: click Upload, choose your CSV or Excel file. The app shows a preview before anything is written — how many rows are new, updates, unchanged, duplicates, warnings or errors, with the exact reason for every problem row and a downloadable error report. Fix your file and re-upload as many times as you need; nothing is written to FormuLab until you click Commit import. If some rows are wrong but others are clean, you can choose to import just the valid ones and skip the rest.

[Screenshot not yet captured: Data Exchange validation preview]

The Import History section shows every attempt — including ones you cancelled or that failed validation — with counts, status, and a link back to the row-level detail. Schema Versions lists every template's current column requirements. An import can never mark something verified or approved on its own — a regulatory rule, a dossier's evidence, a claim, a label, an artwork, a costing override all come in as drafts/unverified, exactly as if you'd typed them in by hand; formal sign-off still happens through that module's own workspace.

Two templates — Stability Protocols and Stability Results — preview and validate normally but cannot yet be committed (a stability study needs a frozen formula/packaging snapshot that a spreadsheet row can't safely provide); importing one reports every row honestly skipped rather than silently doing nothing or faking success.

Full model: DATA_EXCHANGE_CENTER.md and its companion documents (DATA_EXCHANGE_TEMPLATE_REGISTRY.md, DATA_EXCHANGE_IMPORTS.md, DATA_EXCHANGE_EXPORTS.md, DATA_EXCHANGE_VALIDATION.md, DATA_EXCHANGE_SECURITY.md, DATA_EXCHANGE_HISTORY.md, DATA_EXCHANGE_TEMPLATE_CATALOG.md).

24. Reverse Formulation

Open the Reverse Formulation workspace (/reverse-formulation, Formulation group) to benchmark a competitor's product and generate candidate formulas that could plausibly recreate it — a research aid, never a claim of an exact recreation.

- Studies — create a study (code, name, project, product family).
- Products — attach one or more benchmark products to the study; for each, record its ingredient declaration (the label's INCI list, in order — order matters, since INCI order is itself evidence of relative concentration) and any real analytical composition results you have (lab-measured values), each explicitly unverified until a human checks it.
- Mapping — map each declared ingredient to a real material in your own catalog (manual, or a suggested match), then confirm or reject the proposal — a mapping is never auto-accepted.
- Target & constraints — attach a target product profile (pH range, jurisdictions, and more) and a constraint set (required/preferred/ excluded materials, required functions, min/max percentages per material).
- Candidates — generate candidate formulas from the declaration, mappings, target, and constraints (engine/candidateGenerator.ts), each scored (engine/scoringModel.ts) and ranked, with its scoring broken down and explained per dimension, never a single unexplained number. Save a candidate you like, or convert it straight into a real Formulation project version through the normal versioning path — the only module in this guide with that direct bridge (see the composer's own limitation in §0b). A converted version always starts at concept status and is traceable back to its source study/candidate in the audit log; it never inherits approval.

Every line a candidate proposes is attributed model_estimate provenance, the same honest-origin convention every other estimated number in FormuLab uses (§2) — never presented as a measured or confirmed fact.

25. Reports

Open Reports (sidebar, between Approval and Data Exchange) for a single navigation shell over every module's own real export — it does not generate a new document type of its own. From here you can jump straight to a project's Formulation exports, a trial's or study's lab/ stability reports, a dossier's PDF/DOCX, or the Data Exchange Center's current-data downloads, without hunting through each workspace individually.

26. Administration

Open Administration (sidebar, near the bottom) for the organization-wide configuration screens that sit above any one project:

- Materials, suppliers, packaging & factory profiles — the same screens described in §6 and §7, reachable here since they're shared across every project, not owned by one.
- Test definitions — the same catalog described in §17, reachable here for the same reason.
- Links out to Regulatory rules (§20), Approval policies (§5/§12), the Data Exchange Center (§23), and Settings (§28) — Administration is the map between them, not a duplicate of any one.

[Screenshot not yet captured: Administration]

27. Notebooks, Files, and Runs

Three Tools workspaces, useful once your project grows beyond the formula grid itself:

- Notebooks (/notebooks) lists every real .ipynb notebook across every session/project folder, newest first. Create one, or open an existing one, and run cells on the app's local Python kernel — the same file the app itself edits, so there's exactly one copy to keep in sync, never a second "generated" mirror. Open JupyterLab is available once the app-managed Python environment is set up.
- Files (/files) is a real file browser and previewer (image, notebook, spreadsheet, and more, by extension) for your project folder — right-click for the usual file actions.
- Runs (/runs) is the reproducibility log: every command FormuLab itself executed on your behalf (what ran, when, its environment, its inputs/outputs), searchable and filterable, with a Reproduce action that hands the exact original prompt back to you rather than silently re-running something that may no longer be safe to repeat unattended.

[Screenshot not yet captured: Notebooks, Files, and Runs]

28. Settings

Open Settings (bottom of the sidebar) for six sections:

- General — the raw-materials/pricing screen (§6), your workspace folder (where projects/sessions/formulas live on disk — change or reveal it in Explorer/Finder), and app updates (manual check, auto-check toggle, update-badge visibility).
- Models — the provider, model, and API key the composer (§0b) uses to generate a formulation. Never used for anything approval-related — see §30.
- Runtime *(desktop app only)* — the local Python interpreter notebooks and the pipeline run on; auto-detected, with a manual override if you need a specific one.
- Compute — optional remote compute (a remote cluster, or Modal) for work heavier than your own machine.

• Privacy — a data-flow summary: what leaves your machine, to where, and under which model/workspace configuration.
• Appearance — theme (light/warm/dark), UI language (any of the 8 shipped locales), and, in the desktop app, page zoom.

[Screenshot not yet captured: Settings]

29. In-app guide and PDF/DOCX generation

This guide is one Markdown source (docs/USER_GUIDE.md) rendered three ways, never duplicated into a second copy:

• In-app: Help Center -> Open the full guide (or the command palette) renders this exact file through the same MarkdownViewer every other in-app document uses — no separate in-app content model.
• PDF: docs/generated/FormuLab-User-Guide.pdf — a real, paginated document with a cover page, a table of contents with real page numbers, page numbers on every page, embedded screenshots with captions, callout boxes, and readable tables, built with pdf-lib (the same library Dossier PDF export uses, with a guide-specific layout — never the dossier's own content model).
• DOCX: docs/generated/FormuLab-User-Guide.docx — the same content via docx (the same library Dossier DOCX export uses), with a native, Word-refreshable table of contents.

Regenerate both with pnpm docs:guide:generate after editing this file. A screenshot referenced in the Markdown that hasn't been captured yet is skipped with a small "image not yet available" placeholder rather than breaking the build — see docs/PHASE10_SCREENSHOT_MANIFEST.json for which screenshots exist and which are still pending.

30. Roles, safety, and auditability

FormuLab's real role model (ApprovalRole): researcher, chemist, quality, regulatory, production, administrator. There is no login or session-identity system — every workspace that needs a role asks you to select which one you're acting as (an honest limitation, not a security boundary; see §0a's disabled-action explanations for exactly which role each gated action needs).

Gate	Who can act	Enforced
pilot_approved	chemist, quality, administrator	engine-side, not just the UI
production_approved	quality, regulatory, production, admi...	engine-side, not just the UI
Dossier/Claims formal review or export	regulatory, quality, administrator	engine-side, not just the UI
Laboratory method assignment/promotion	chemist, quality, administrator	engine-side, not just the UI

Note: No automated actor can ever approve. An AI-generated candidate, an imported spreadsheet row, or a system process can never set pilot_approved/production_approved, record a regulatory review, or verify evidence — only a named human, through that module's own workspace. This is checked in the engine layer itself (packages/shared/src/engine/), not merely hidden in the interface — see FORMULA_VERSIONING.md.

Every meaningful change — a saved version, an approval, a status change, an evidence upload, a review, an export — appends to that project's own audit log; nothing in FormuLab silently rewrites history. A number this guide cannot honestly compute stays labelled missing / unknown / not_applicable / not_verified / estimated rather than defaulting to zero or an invented value — the same discipline every module above follows, not a one-off caveat.

Known limitations

For how the sidebar's workspaces are organized and why, see INFORMATION_ARCHITECTURE.md, WORKSPACES.md and NAVIGATION_AND_CONTEXT.md.

See IMPLEMENTATION_STATUS.md for the authoritative list of what is built versus not yet started. In short: the Kenya/EAC regulatory engine (§20 above) is implemented, including version-bound human review, persisted evidence confirmations, multi-jurisdiction approval readiness, and rule source verification. Phase 3 (regulatory dossiers, evidence matrix, /dossiers) and Phase 4 (product claims, labels, artwork, formula-label consistency, /claims-labels) are both implemented, including their workspace UIs — see REGULATORY_DOSSIERS.md, PRODUCT_CLAIMS.md, PRODUCT_LABELS.md. Dossiers generates a real, watermarked final PDF/DOCX (Phase 7 — see §21a); Claims & Labels does not have its own equivalent document export yet. Phase 5 (Design of Experiments, /doe) is implemented, including its workspace UI — see DESIGN_OF_EXPERIMENTS.md; within it, definitive_screening/mixture_simplex_centroid designs, fractions beyond a half-fraction, and Plackett-Burman sizes beyond N=12 are refused rather than faked, and an analysis-results export can never be re-imported as a native analysis. Reverse Formulation (§24) is implemented, including candidate generation, scoring, and a direct conversion path into a real Formulation project version. Laboratory trials and stability studies (§16–19 above) are implemented, but automatic shelf-life prediction is deliberately not — see STABILITY_TRENDS.md. The Advanced Optimizer's screen has no builder for composition, ratio or conditional constraints (only functional-group constraints, property targets, a cost ceiling, and the automatic compatibility/safety exclusion are user-facing) and no lexicographic-priority selector; the Substitution dialog's system mode does not yet wire graded compatibility/ safety risk into a system's score (real hard exclusions still apply) — see ADVANCED_OPTIMIZER.md and SYSTEM_SUBSTITUTION.md. Compatibility and safety are deterministic rule engines against a hand-maintained, explicitly non-exhaustive seed rule set — they are not a regulatory engine and do not establish legal compliance. The Approval workspace calls approval readiness (including the lab/stability policies in LAB_STABILITY_APPROVAL.md) for real — see APPROVAL_WORKFLOW.md. Its approval-policy editor supports full field editing, product-family/packaging-SKU scoping, clone, retire and revision history/restore, plus deterministic conflict resolution when more than one active policy matches — see APPROVAL_POLICIES.md for the narrower gaps that remain. No native GUI click-through of the packaged app exists yet in this environment — see APPROVAL_MANUAL_SMOKE_TEST.md. Nothing in this guide describes an unimplemented module as available.

This guide now references real screenshot filenames throughout (the <module>-<page>-<state>-<theme>-<locale>.png convention from docs/PHASE10_SCREENSHOT_MANIFEST.json), but no screenshot has actually been captured yet — every reference currently resolves to a missing image, rendered as a small, honest placeholder in the PDF/DOCX rather than a broken image or a silently-dropped one. This is a genuine, disclosed gap, not an oversight: capturing a real screenshot requires driving the actual Tauri desktop window (launching it against the Session 5 documentation fixture, navigating it, and taking a pixel capture), and no reliable, safe automation driver for that native window was available in this session's environment — attempting it with fragile timing-based input injection risked capturing and shipping a wrong or half-loaded state, which would be worse than no image. The safe, deterministic documentation fixture and screenshot manifest exist and are fully wired up (every fixture record is synthetic and clearly marked DEMO-, seeded into an isolated profile that is never the app's real data folder); only the actual capture pass remains, for a future session with a proper window-automation harness.